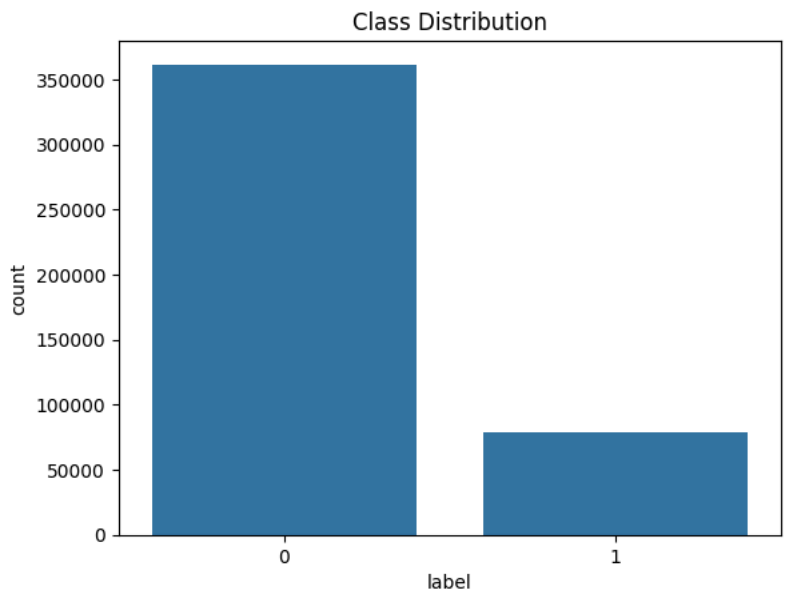
```
import pandas as pd
import numpy as np
import re
import nltk
import matplotlib.pyplot as plt
import seaborn as sns

from nltk.corpus import stopwords

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.linear_model import LogisticRegression
from sklearn.svm import LinearSVC
from sklearn.naive_bayes import MultinomialNB
from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score, f1_score, classification_report, confusion_matrix
```

```
df = pd.read_csv("HateSpeechDataset.csv")

df = df[['Content', 'Label']]
df.rename(columns={'Content': 'text', 'Label': 'label'}, inplace=True)

# 🔥 FIX DATASET ISSUE
df['label'] = pd.to_numeric(df['label'], errors='coerce')
df.dropna(inplace=True)
df['label'] = df['label'].astype(int)

df.reset_index(drop=True, inplace=True)

print(df.head())
print(df['label'].value_counts())
```

```
                                                text  label
0  denial of normal the con be asked to comment o...      1
1  just by being able to tweet this insufferable ...      1
2  that is retarded you too cute to be single tha...      1
3  thought of a real badass mongol style declarat...      1
4                                afro american basho      1
label
0    361594
1     79305
Name: count, dtype: int64
```

```
sns.countplot(x='label', data=df)
plt.title("Class Distribution")
plt.show()
```




```
nltk.download('stopwords')
stop_words = set(stopwords.words('english'))

def clean_text(text):
    text = str(text).lower()
```

```
    text = re.sub(r'http\S+', '', text)
    text = re.sub(r'@[\w]*', '', text)
    text = re.sub(r'#', '', text)

    text = re.sub(r'[^a-zA-Z\s]', '', text)

    words = text.split()
    words = [w for w in words if w not in stop_words and len(w) > 2]

    return " ".join(words)

df['clean_text'] = df['text'].apply(clean_text)

df[['text', 'clean_text']].head()
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

	text	clean_text
0	denial of normal the con be asked to comment o...	denial normal con asked comment tragedies emot...
1	just by being able to tweet this insufferable ...	able tweet insufferable bullshit proves trump ...
2	that is retarded you too cute to be single tha...	retarded cute single life
3	thought of a real badass mongol style declarat...	thought real badass mongol style declaration w...
4	afro american basho	afro american basho

```
from wordcloud import WordCloud

text = " ".join(df['clean_text'])

wc = WordCloud(width=800, height=400, background_color='white').generate(text)

plt.imshow(wc)
plt.axis('off')
plt.title("WordCloud")
plt.show()
```

```
tfidf = TfidfVectorizer(
    max_features=10000,
    ngram_range=(1,2),
    min_df=2,
    max_df=0.9
)

X = tfidf.fit_transform(df['clean_text'])
y = df['label']
```

```
X_train, X_val, y_train, y_val = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
models = {
    "Logistic Regression": LogisticRegression(max_iter=1000, class_weight='balanced'),
    "Linear SVM": LinearSVC(class_weight='balanced'),
    "Naive Bayes": MultinomialNB(),
    "Random Forest": RandomForestClassifier(n_estimators=50,
    max_depth=20,
    n_jobs=-1 )
}
```

```
results = []

for name, model in models.items():
    model.fit(X_train, y_train)
    preds = model.predict(X_val)

    acc = accuracy_score(y_val, preds)
    f1 = f1_score(y_val, preds)

    results.append([name, acc, f1])

    print(name)
    print("Accuracy:", acc)
    print("F1 Score:", f1)
    print("-"*30)
```

```
Logistic Regression
Accuracy: 0.8199364935359492
F1 Score: 0.6150974498206148
------------------------------
Linear SVM
Accuracy: 0.8167384894533908
F1 Score: 0.6098691516585389
------------------------------
Naive Bayes
Accuracy: 0.8563393059650715
F1 Score: 0.5430344131015078
------------------------------
Random Forest
Accuracy: 0.8211839419369471
F1 Score: 0.008925204274041483
------------------------------
```

```
results_df = pd.DataFrame(results, columns=["Model", "Accuracy", "F1 Score"])
print(results_df)
```

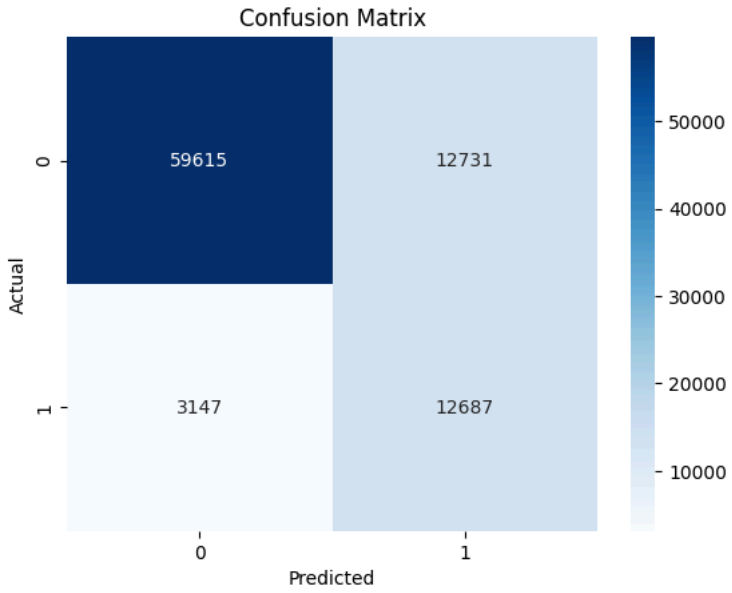
```
                 Model  Accuracy  F1 Score
0  Logistic Regression  0.819936  0.615097
1           Linear SVM  0.816738  0.609869
2          Naive Bayes  0.856339  0.543034
3        Random Forest  0.821184  0.008925
```

```
best_model = LogisticRegression(class_weight='balanced')
best_model.fit(X_train, y_train)

y_pred = best_model.predict(X_val)
```

```
print("Accuracy:", accuracy_score(y_val, y_pred))

print("\nClassification Report:\n")
print(classification_report(y_val, y_pred))
```

```
Accuracy: 0.8199364935359492

Classification Report:

              precision    recall  f1-score   support

           0       0.95      0.82      0.88     72346
           1       0.50      0.80      0.62     15834

    accuracy                           0.82     88180
   macro avg       0.72      0.81      0.75     88180
weighted avg       0.87      0.82      0.83     88180
```

```
cm = confusion_matrix(y_val, y_pred)

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```




```
test_samples = [
    "you are stupid",
    "i love this product",
    "go kill yourself",
    "have a nice day"
]

for text in test_samples:
    cleaned = clean_text(text)
    vector = tfidf.transform([cleaned])
    pred = best_model.predict(vector)[0]

    print(text, "→", "Toxic" if pred==1 else "Non-Toxic")
```

```
you are stupid → Toxic
i love this product → Non-Toxic
go kill yourself → Toxic
have a nice day → Non-Toxic
```

```
import pickle

pickle.dump(best_model, open("model.pkl", "wb"))
pickle.dump(tfidf, open("tfidf.pkl", "wb"))
```

Start coding or generate with AI.